INSTITUTE OF INNOVATION

Sample Paper Mid-Semester Examination

Computer Organization and Architecture(COA)

Total Marks: 50
Duration: 1 Hours 30 mins
Total Questions: 33 (MCQs+Subjective)

Instructions:

- Section A: 25 questions × 1 mark = 25 marks
- Section B : 5 questions (Attempt any 3) × 5 marks = 15 marks
- Section C : 3 questions (Attempt any 1) × 10 marks = 10 marks
- Use of calculator is allowed
- Read all questions carefully before answering.
- Attempt only the required number of questions in each section

Section -A- MCQs (25 × 1marks)

Q1. Which gate outputs 1 only if *at least one* input is 1?
A) AND B) OR C) NOR D) NOT

Q2. XOR gate gives output 1 when:
A) Inputs are same B) Inputs differ C) Both inputs are 0 D) Both inputs are 1

Q3. Full adder has how many inputs?
A) 2 B) 3 C) 4 D) 1

Q4. A demultiplexer performs the function of:
A) Many-to-one B) One-to-many C) Binary-to-decimal D) Memory expansion

Q5. A 2-bit comparator checks:
A) Only MSB B) Only LSB C) All bits simultaneously D) Two bits at a time

Q6. An SR flip-flop invalid state occurs when:
A) S=0, R=1 B) S=1, R=1 C) S=0, R=0 D) S=1, R=0

Q7. Asynchronous counters are also called:
A) Parallel counters B) Ripple counters C) Up counters D) Mod counters

Q8. A shift register moves data:
A) Upward B) Downward C) Left or right D) Randomly

Q9. Von Neumann architecture uses:
A) Separate bus for data & instructions
B) Single bus for data & instructions
C) No control unit
D) Harvard buses

Q**10**. Which flip-flop changes its output only at the moment of a clock transition (rising or falling edge)?
A) SR flip-flop
B) Level-triggered latch

C) Edge-triggered flip-flop
D) Asynchronous flip-flop

Q11. The borrow output in a full subtractor is given by:
A) B'A + BC   B) B + C_in   C) B'C + AC'   D) B'C'

Q12. A 4:1 multiplexer requires how many select lines?
A) 1   B) 2   C) 3   D) 4

Q13. The output of an SR flip-flop when S=1 and R=0 is:
A) Toggle   B) Reset   C) Set   D) Undefined

Q14. In a synchronous counter, all flip-flops:
A) Trigger one after another
B) Trigger at different edges
C) Trigger at same clock edge
D) Are combinational devices

Q15. Circular shift left (CSL) moves:
A) MSB to LSB
B) LSB to MSB
C) Both A and B
D) None

Q16. A 3x8 decoder requires how many enable inputs?
A) 0   B) 1   C) 2   D) 3

Q17. In 2's complement subtraction A – B, the operation performed is:
A) A + B
B) A + B' + 1

C) A – B – 1
D) B – A

Q18. In which addressing mode is the effective address obtained by adding a constant value to the contents of an index register?

A) Direct addressing

B) Immediate addressing

C) Indexed addressing

D) Register indirect addressing

Q19. The ALU receives control signals from the:
A) Program counter
B) Control unit
C) IR register
D) Memory data register

Q**20.** An asynchronous counter has:

A) All flip-flops triggered together
B) Clock applied only to the first flip-flop
C) No flip-flops
D) Synchronous enable signal

Q**21.** A 3-bit **asynchronous up counter** has its flip-flops named Q2 (MSB), Q1, and Q0 (LSB).
Which flip-flop receives the clock directly from the external clock source?

A) Q2 only
B) Q1 only
C) Q0 only
D) All flip-flops receive the same clock

Q22. In a 4×1 multiplexer, implementing a Boolean function without simplification is known as:
A) K-map implementation
B) MUX realization
C) SOP reduction
D) Encoder

Q**23.** A demultiplexer performs the reverse operation of:

A) Encoder
B) Decoder
C) Multiplexer
D) Register

Q**24.** A 4-bit **adder–subtractor circuit** uses XOR gates on the B inputs and a **Mode** control (M), where:

- M = 0 → Addition
- M = 1 → Subtraction

Which statement is TRUE about how subtraction is performed?

A) B is inverted and Cin = 0
B) B is passed unchanged and Cin = 1
C) B is XORed with M and Cin = M
D) B is complemented and Cin = 1 always

Q**25.** A 1-to-8 demultiplexer has inputs: Data input D, select lines S2 S1 S0, and outputs Y0–Y7.
If D = 1 and S2S1S0 = 101, which output will be HIGH (1)?

A) Y1
B) Y3
C) Y5
D) Y7

SECTION–B (3 × 5marks)

Q1. Why are NAND and NOR gates called *universal gates*? Explain how any basic gate can be implemented using these universal gates

Q2. Compare asynchronous and synchronous counters. Highlight any two differences and explain which type avoids propagation delays.

Q3. What is the difference between Von Neumann and Harvard architectures? State any two advantages of Harvard architecture.

Q4. Design a 4-bit adder–subtractor.

Q5. Design a 3-bit asynchronous down counter.

SECTION - C (1 × 10marks)

Q1. Define and explain the three major categories of microoperations performed in a computer system:

- Arithmetic microoperations

- Logic microoperations
- Shift microoperations

For each category, describe at least three operations with a short explanation of their purpose.

Q2. Explain the working principle of a PISO (Parallel-In Serial-Out) register.

Q3. Design a 3-to-8 decoder using 1-to-2 decoders.